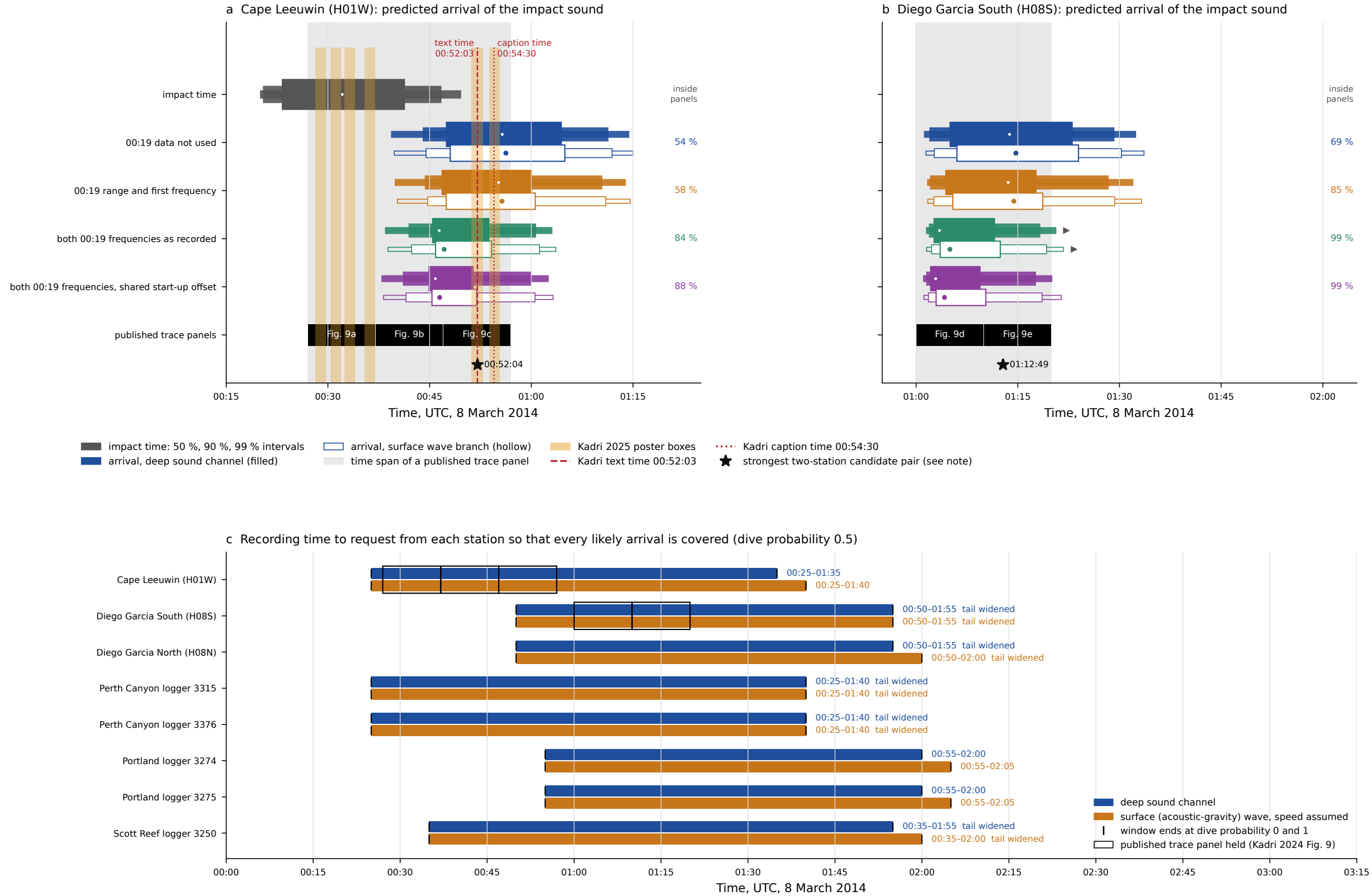
Predicted arrival of the impact sound at Cape Leeuwin and Diego Garcia South against the published traces, and the recording time to request



Plain note: Panels a and b show when the sound of the impact would arrive at Cape Leeuwin and at Diego Garcia South, for four ways to use the last satellite data at 00:19. The grey bar is the time of the impact. The black bars are the 10-minute trace panels that Kadri published; the percentage at the right is the part of each predicted arrival inside those panels. The orange areas are the six boxes on Kadri's 2025 poster. The red lines are the two times Kadri gives for his signal. The star marks the strongest pair of candidate signals found tonight. Panel c gives the recording time to request from each station. No detection is claimed.

Technical note: Source posterior: end-of-flight run J (74 seeds, 313.6 M rows) + northern supplement; p (dive) = 0.5; weights = end-of-flight two-factor table total, C2 silent (q = 1/3), survival factor, strata by p_core x Zhat (run_j_windows.py, prereg bc99b07). Arrival = impact time + WGS84 geodesic / c; deep sound channel c ~ N(1.482, 0.006) km/s; acoustic-gravity branch c ~ U(1.40, 1.50) km/s (assumed, not modelled); no refraction or blockage. Bars: 50 % (thick), 90 %, 99 % (thin) intervals, dot = median; arrow = upper tail not converged across seeds (> 5 min spread). Kish ESS at Cape Leeuwin: D0′ (held out) 1,336,700, D2 (R600 BTO + BFO) 107,226, D3 (both BFOs raw) 5,671, D4 (both BFOs + shared start-up offset) 25,414. Rows are the 00:19 options (D0′, D2, D3, D4); the log-on cause is mixed by the end-of-flight weights (the earlier chart's 'other / fuel exhaustion' and 'alive / silent' arms are not separate in run J). D1 and D5 are not in the run J windows design (gap). Kadri 2024 Fig. 9 panels a–c 00:27–00:57 (H01W), d–e 01:00–01:20 (H08S), digitised. Kadri times: text time 00:52:03 (p. 9) and caption time 00:54:30 (Table 1). Poster boxes: SnT 2025 panels (a) and (c), digitised. Strongest two-station pair (searches note): H01W 00:52:04 (envelope matched filter, p 0.043) x H08S 01:12:49 (p 0.039; airgun-suppressed residual p_B 0.30), lag 1,245 s inside the predicted 809–1,536 s; 19 lag-consistent pairs vs 8.8 expected, permutation p 0.019 over 36 receiver-pair tests. BEARING CAVEAT: the pair's arc locus (33.6 S 94.5 E) lies at 268.9 deg from H01W and 144.8 deg from H08S; Kadri gives 57 deg (text p. 9) or 306.18 deg (Table 1 caption) for the 00:52 signal, neither within 37 deg of 268.9; the nearest Table 1 bearing is 00:52:50 at 274.84 deg, 46 s later. Request windows: union over options with ESS >= 1,000 of [q0.5 − 10 min, q99.5 + 20 min]. PREDICTIVE, NOT CALIBRATED. C1: propagation slope misfit +10.4 / +7.4 dB per octave at H01W / H08S (after the EW0112 source result a source-term caveat; air9 verdict NOT EXPLAINED, gap G23). C2: coupling efficiency eta not calibrated; swept over four decades; F-35A upper anchor eta = 2.1e-4 is provisional (from Brown 2026's quote of Metz 2023; re-derivation from Metz pending). No detection is claimed.